



**CELEBAL**  
TECHNOLOGIES

**Celebal Summer Internship 2026**

**WEEK 3 – ASSIGNMENT**

# Sales Data Analysis Using SQL

## Objective

The objective of this project is to analyze the Superstore sales dataset using SQL. The analysis uses Subqueries, Common Table Expressions (CTEs), Window Functions, and JOIN operations to answer business questions and identify important sales trends.

## Step 1: Load Superstore dataset into a table (superstore\_raw).

### Load the Dataset

The Superstore dataset was imported into a table named superstore\_raw.



```
1 • select count(*) AS Total_Rows
2 FROM superstore_raw;
```

## Output



| Total_Rows |
|------------|
| 9694       |

## Insight

The dataset contains 9694 sales records which will be used for analysis.

**Step 2: Create tables (customers, orders, products) from the dataset. Insert data into these tables using SELECT DISTINCT**

### Create Customers Table

A separate customers table was created using distinct customer information from the raw dataset.

```
CREATE TABLE customers AS  
SELECT DISTINCT  
  `Customer ID`,  
  `Customer Name`,  
  Segment  
FROM superstore_raw;
```

### Verification

```
SELECT COUNT(*) FROM customers;
```

### Output

| Result Grid |          | Filter |
|-------------|----------|--------|
|             | COUNT(*) |        |
| ▶           | 793      |        |

## Insight

The dataset contains 793 unique customers.

## Create Products Table

A products table was created to store unique product details.

```
CREATE TABLE products AS  
SELECT DISTINCT  
    `Product ID`,  
    `Product Name`,  
    Category,  
    `Sub-Category`  
FROM superstore_raw;
```

## Verification

```
SELECT COUNT(*) FROM products;
```

## Output



|   | COUNT(*) |
|---|----------|
| ▶ | 1842     |

## Insight

The company sells 1862 unique products.

## Create Orders Table

The orders table stores transaction details such as sales, quantity, profit, and order date.

```
CREATE TABLE orders AS
SELECT distinct
    `Order ID`,
    `Customer ID`,
    `Product ID`,
    Sales,
    Quantity,
    Profit,
    `Order Date`,
    Region
FROM superstore_raw;
```

## Verification

```
SELECT COUNT(*) FROM orders;
```

## Output



The screenshot shows a database interface with a 'Result Grid' tab. It displays the output of a SQL query. The first row shows the column 'COUNT(\*)' and the second row shows the value '9694'. There is also a 'Filter Rows' button and a search input field.

|   | COUNT(*) |
|---|----------|
| ▶ | 9694     |

## Insight

The orders table contains all transaction records required for sales analysis.

## Step 3: Apply subqueries to filter data (above average sales, highest order per customer).

### Find Orders Above Average Sales (Subquery)

This query identifies orders whose sales are higher than the average sales value.

```
SELECT *  
FROM orders  
WHERE Sales >  
(  
    SELECT AVG(Sales)  
    FROM orders  
);
```

### Output

| Order ID       | Customer ID | Product ID      | Sales    | Quantity | Profit     | Order Date | Region  |
|----------------|-------------|-----------------|----------|----------|------------|------------|---------|
| CA-2016-152156 | CG-12520    | FUR-BO-10001798 | 261.96   | 2        | 41.9136    | 11/8/2016  | South   |
| CA-2016-152156 | CG-12520    | FUR-CH-10000454 | 731.94   | 3        | 219.582    | 11/8/2016  | South   |
| US-2015-108966 | SO-20335    | FUR-TA-10000577 | 957.5775 | 5        | -383.031   | 10/11/2015 | South   |
| CA-2014-115812 | BH-11710    | TEC-PH-10002275 | 907.152  | 6        | 90.7152    | 6/9/2014   | West    |
| CA-2014-115812 | BH-11710    | FUR-TA-10001539 | 1706.184 | 9        | 85.3092    | 6/9/2014   | West    |
| CA-2014-115812 | BH-11710    | TEC-PH-10002033 | 911.424  | 4        | 68.3568    | 6/9/2014   | West    |
| CA-2016-161389 | IM-15070    | OFF-BI-10003656 | 407.976  | 3        | 132.5922   | 12/5/2016  | West    |
| CA-2014-105893 | PK-19075    | OFF-ST-10004186 | 665.88   | 6        | 13.3176    | 11/11/2014 | Central |
| CA-2015-106320 | EB-13870    | FUR-TA-10000577 | 1044.63  | 3        | 240.2649   | 9/25/2015  | West    |
| US-2015-150630 | TB-21520    | FUR-BO-10004834 | 3083.43  | 7        | -1665.0522 | 9/17/2015  | East    |
| CA-2016-117590 | GH-14485    | TEC-PH-10004977 | 1097.544 | 7        | 123.4737   | 12/8/2016  | Central |
| CA-2015-117415 | SN-20710    | FUR-BO-10002545 | 532.3992 | 3        | -46.9764   | 12/27/2015 | Central |
| CA-2015-117415 | SN-20710    | TEC-PH-10000486 | 371.168  | 4        | 41.7564    | 12/27/2015 | Central |
| CA-2016-105816 | JM-15265    | TEC-PH-10002447 | 1029.95  | 5        | 298.6855   | 12/11/2016 | East    |
| CA-2016-111682 | TB-21055    | FUR-CH-10003968 | 319.41   | 5        | 7.098      | 6/17/2016  | East    |
| CA-2014-106376 | BS-11590    | OFF-AR-10002671 | 1113.024 | 8        | 111.3024   | 12/5/2014  | West    |
| US-2015-134026 | JE-15745    | FUR-CH-10000513 | 831.936  | 8        | -114.3912  | 4/26/2015  | South   |
| CA-2017-140088 | PO-18865    | FUR-CH-10000863 | 301.96   | 2        | 33.2156    | 5/28/2017  | South   |
| US-2015-156867 | LC-16870    | TEC-AC-10001552 | 238.896  | 6        | -26.8758   | 11/13/2015 | West    |
| CA-2015-129476 | PA-19060    | TEC-AC-10000844 | 339.96   | 5        | 67.992     | 10/15/2015 | Central |
| CA-2015-110457 | DK-13090    | FUR-TA-10001768 | 787.53   | 3        | 165.3813   | 3/2/2015   | West    |
| US-2014-152030 | AD-10180    | FUR-CH-10004063 | 600.558  | 3        | -8.5794    | 12/26/2014 | Central |
| US-2014-134614 | PF-19165    | FUR-TA-10004534 | 617.7    | 6        | -407.682   | 9/20/2014  | Central |

### Insight

These orders contribute more revenue than the average order and are considered high-value transactions.

### **Find Highest Order Per Customer (Subquery)**

This query identifies the highest-value order placed by each customer.

```
SELECT *  
FROM orders o  
WHERE Sales =  
(  
    SELECT MAX(Sales)  
    FROM orders  
    WHERE `Customer ID` = o.`Customer ID`  
);
```

Output



| Result Grid |                |              |                 |          |                    |            |            |         |
|-------------|----------------|--------------|-----------------|----------|--------------------|------------|------------|---------|
|             |                | Filter Rows: | Export:         |          | Wrap Cell Content: |            |            |         |
|             | Order ID       | Customer ID  | Product ID      | Sales    | Quantity           | Profit     | Order Date | Region  |
| ▶           | CA-2016-152156 | CG-12520     | FUR-CH-10000454 | 731.94   | 3                  | 219.582    | 11/8/2016  | South   |
|             | US-2015-108966 | SO-20335     | FUR-TA-10000577 | 957.5775 | 5                  | -383.031   | 10/11/2015 | South   |
|             | CA-2014-115812 | BH-11710     | FUR-TA-10001539 | 1706.184 | 9                  | 85.3092    | 6/9/2014   | West    |
|             | CA-2015-106320 | EB-13870     | FUR-TA-10000577 | 1044.63  | 3                  | 240.2649   | 9/25/2015  | West    |
|             | US-2015-150630 | TB-21520     | FUR-BO-10004834 | 3083.43  | 7                  | -1665.0522 | 9/17/2015  | East    |
|             | CA-2016-117590 | GH-14485     | TEC-PH-10004977 | 1097.544 | 7                  | 123.4737   | 12/8/2016  | Central |
|             | CA-2015-117415 | SN-20710     | FUR-BO-10002545 | 532.3992 | 3                  | -46.9764   | 12/27/2015 | Central |
|             | CA-2016-105816 | JM-15265     | TEC-PH-10002447 | 1029.95  | 5                  | 298.6855   | 12/11/2016 | East    |
|             | CA-2016-111682 | TB-21055     | FUR-CH-10003968 | 319.41   | 5                  | 7.098      | 6/17/2016  | East    |
|             | CA-2014-106376 | BS-11590     | OFF-AR-10002671 | 1113.024 | 8                  | 111.3024   | 12/5/2014  | West    |
|             | CA-2015-129476 | PA-19060     | TEC-AC-10000844 | 339.96   | 5                  | 67.992     | 10/15/2015 | Central |
|             | CA-2015-110744 | HA-14920     | OFF-ST-10003656 | 671.93   | 7                  | 20.1579    | 9/7/2015   | West    |
|             | CA-2016-114489 | JE-16165     | FUR-CH-10000454 | 1951.84  | 8                  | 585.552    | 12/5/2016  | Central |
|             | CA-2014-139892 | BM-11140     | TEC-MA-10000822 | 8159.952 | 8                  | -1359.992  | 9/8/2014   | Central |
|             | CA-2014-118962 | CS-12130     | FUR-CH-10003817 | 340.144  | 7                  | 21.259     | 8/5/2014   | West    |
|             | CA-2014-166191 | DK-13150     | TEC-AC-10004659 | 408.744  | 7                  | 76.6395    | 12/5/2014  | Central |
|             | CA-2015-102281 | MP-17470     | FUR-BO-10002613 | 899.136  | 4                  | 112.392    | 10/12/2015 | East    |
|             | CA-2016-130162 | JH-15910     | TEC-PH-10002563 | 302.376  | 3                  | 22.6782    | 10/28/2016 | West    |
|             | CA-2015-169397 | JB-15925     | TEC-MA-10001148 | 479.988  | 4                  | -383.9904  | 12/24/2015 | East    |
|             | CA-2015-163055 | DS-13180     | FUR-TA-10003748 | 622.45   | 5                  | 136.939    | 8/9/2015   | Central |
|             | CA-2014-131926 | DW-13480     | FUR-CH-10004063 | 2001.86  | 7                  | 580.5394   | 6/1/2014   | Central |
|             | CA-2016-145625 | KC-16540     | TEC-AC-10003832 | 3347.37  | 13                 | 636.0003   | 9/11/2016  | West    |
|             | CA-2016-146941 | DL-13315     | OFF-EN-10003296 | 361.92   | 4                  | 162.864    | 12/10/2016 | East    |
|             | US-2014-106992 | SB-20290     | TEC-MA-10000822 | 3059.982 | 3                  | -509.997   | 9/19/2014  | Central |
|             | CA-2016-125318 | RC-19825     | TEC-PH-10001433 | 328.224  | 4                  | 28.7196    | 6/6/2016   | Central |

## Insight

This helps understand the maximum spending capacity of each customer.

## Step 4: Use CTEs to compute aggregations (total sales per customer).

A CTE is used to calculate the total sales generated by each customer.



```
WITH customer_sales AS
(
    SELECT
        `Customer ID`,
        SUM(Sales) AS Total_Sales
    FROM orders
    GROUP BY `Customer ID`
)
SELECT *
FROM customer_sales
ORDER BY Total_Sales DESC;
```

Output

| Result Grid    Filter Rows: <input type="text"/>   Export: |             |                    |
|---------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|
|                                                                                                                                             | Customer ID | Total_Sales        |
| ▶                                                                                                                                           | SM-20320    | 25043.05           |
|                                                                                                                                             | TC-20980    | 19017.847999999998 |
|                                                                                                                                             | RB-19360    | 15117.339          |
|                                                                                                                                             | TA-21385    | 14595.62           |
|                                                                                                                                             | AB-10105    | 14355.610999999997 |
|                                                                                                                                             | SC-20095    | 14142.333999999999 |
|                                                                                                                                             | KL-16645    | 14071.917          |
|                                                                                                                                             | HL-15040    | 12873.297999999999 |
|                                                                                                                                             | SE-20110    | 12209.438000000002 |
|                                                                                                                                             | CC-12370    | 12129.072          |
|                                                                                                                                             | TS-21370    | 11885.871          |
|                                                                                                                                             | GT-14710    | 11820.119999999997 |
|                                                                                                                                             | BM-11140    | 11609.900000000001 |
|                                                                                                                                             | SV-20365    | 11420.645999999997 |
|                                                                                                                                             | CJ-12010    | 11079.742          |
|                                                                                                                                             | CL-12565    | 10880.545999999998 |
|                                                                                                                                             | ME-17320    | 10635.917999999998 |
|                                                                                                                                             | KF-16285    | 10604.266          |
|                                                                                                                                             | BS-11365    | 10501.652999999998 |
|                                                                                                                                             | EH-13765    | 10005.980000000001 |
|                                                                                                                                             | JL-15835    | 9403.003           |
|                                                                                                                                             | GT-14635    | 9351.211999999998  |
|                                                                                                                                             | HW-14935    | 9300.253999999997  |

Result 12 × 

## Insight

This query helps identify customers who generate the highest revenue.

**Step 5: Apply window functions (ROW\_NUMBER, RANK) for ranking and analysis.**

## Assign Row Numbers to Orders (Window Function)

Assigns a sequence number to each order within a customer.

```
SELECT *  
FROM customer_sales  
ORDER BY Total_Sales DESC;  
SELECT  
    `Customer ID`,  
    Sales,  
    ROW_NUMBER() OVER  
    (  
        PARTITION BY `Customer ID`  
        ORDER BY Sales DESC  
    ) AS Row_Num  
FROM orders;
```

Output

| Result Grid  |             |          |         |
|--------------|-------------|----------|---------|
| Filter Rows: |             |          |         |
|              | Customer ID | Sales    | Row_Num |
| ▶            | AA-10315    | 3930.072 | 1       |
|              | AA-10315    | 673.568  | 2       |
|              | AA-10315    | 431.976  | 3       |
|              | AA-10315    | 362.94   | 4       |
|              | AA-10315    | 52.98    | 5       |
|              | AA-10315    | 41.72    | 6       |
|              | AA-10315    | 26.96    | 7       |
|              | AA-10315    | 14.94    | 8       |
|              | AA-10315    | 14.56    | 9       |
|              | AA-10315    | 11.54    | 10      |
|              | AA-10315    | 2.304    | 11      |
|              | AA-10375    | 499.98   | 1       |
|              | AA-10375    | 149.97   | 2       |
|              | AA-10375    | 116.98   | 3       |
|              | AA-10375    | 84.96    | 4       |
|              | AA-10375    | 39.96    | 5       |
|              | AA-10375    | 34.272   | 6       |
|              | AA-10375    | 28.4     | 7       |
|              | AA-10375    | 17.94    | 8       |
|              | AA-10375    | 16.9     | 9       |
|              | AA-10375    | 16.768   | 10      |
|              | AA-10375    | 16.52    | 11      |
|              | AA-10375    | 14.952   | 12      |
|              | AA-10375    | 8.26     | 13      |
|              | AA-10375    | 5.78     | 14      |

Insight

This helps analyze customer purchasing patterns.

## Rank Customers Based on Total Sales (Window Function)

Customers are ranked according to their total sales.

```
SELECT
    `Customer ID`,
    SUM(Sales) AS Total_Sales,
    RANK() OVER
    (
        ORDER BY SUM(Sales) DESC
    ) AS Customer_Rank
FROM orders
GROUP BY `Customer ID`;
```

## Output

| Result Grid     Filter Rows: <input type="text"/>   Exp |             |                    |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|---------------|
|                                                                                                                                                                                                                           | Customer ID | Total_Sales        | Customer_Rank |
| ▶                                                                                                                                                                                                                         | SM-20320    | 25043.05           | 1             |
|                                                                                                                                                                                                                           | TC-20980    | 19017.847999999998 | 2             |
|                                                                                                                                                                                                                           | RB-19360    | 15117.339          | 3             |
|                                                                                                                                                                                                                           | TA-21385    | 14595.62           | 4             |
|                                                                                                                                                                                                                           | AB-10105    | 14355.610999999997 | 5             |
|                                                                                                                                                                                                                           | SC-20095    | 14142.333999999999 | 6             |
|                                                                                                                                                                                                                           | KL-16645    | 14071.917          | 7             |
|                                                                                                                                                                                                                           | HL-15040    | 12873.297999999999 | 8             |
|                                                                                                                                                                                                                           | SE-20110    | 12209.438000000002 | 9             |
|                                                                                                                                                                                                                           | CC-12370    | 12129.072          | 10            |
|                                                                                                                                                                                                                           | TS-21370    | 11885.871          | 11            |
|                                                                                                                                                                                                                           | GT-14710    | 11820.119999999997 | 12            |
|                                                                                                                                                                                                                           | BM-11140    | 11609.900000000001 | 13            |
|                                                                                                                                                                                                                           | SV-20365    | 11420.645999999997 | 14            |
|                                                                                                                                                                                                                           | CJ-12010    | 11079.742          | 15            |
|                                                                                                                                                                                                                           | CL-12565    | 10880.545999999998 | 16            |
|                                                                                                                                                                                                                           | ME-17320    | 10635.917999999998 | 17            |
|                                                                                                                                                                                                                           | KF-16285    | 10604.266          | 18            |
|                                                                                                                                                                                                                           | BS-11365    | 10501.652999999998 | 19            |
|                                                                                                                                                                                                                           | EH-13765    | 10005.980000000001 | 20            |
|                                                                                                                                                                                                                           | JL-15835    | 9403.003           | 21            |
|                                                                                                                                                                                                                           | CT-14635    | 9351.711000000000  | 22            |

## Insight

Ranking helps identify the best-performing customers.

**Step 6: Combine JOIN + CTE + Window Functions for final result (customer, total sales, rank).**

Combines customer information, total sales, and ranking into a single report.

```
WITH customer_sales AS
(
    SELECT
        `Customer ID`,
        SUM(Sales) AS Total_Sales
    FROM orders
    GROUP BY `Customer ID`
)
SELECT
    c.`Customer ID`,
    c.`Customer Name`,
    cs.Total_Sales,
    RANK() OVER (
        ORDER BY cs.Total_Sales DESC) AS Customer_Rank
FROM customers c
JOIN customer_sales cs
ON c.`Customer ID` = cs.`Customer ID`;
```

Output



| Result Grid    Filter Rows: <input type="text"/>   Export:    Wrap Cell Content:  |             |                    |                    |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|--------------------|---------------|
|                                                                                                                                                                                                                                                                                                                                         | Customer ID | Customer Name      | Total_Sales        | Customer_Rank |
| ▶                                                                                                                                                                                                                                                                                                                                       | SM-20320    | Sean Miller        | 25043.05           | 1             |
|                                                                                                                                                                                                                                                                                                                                         | TC-20980    | Tamara Chand       | 19017.847999999998 | 2             |
|                                                                                                                                                                                                                                                                                                                                         | RB-19360    | Raymond Buch       | 15117.339          | 3             |
|                                                                                                                                                                                                                                                                                                                                         | TA-21385    | Tom Ashbrook       | 14595.62           | 4             |
|                                                                                                                                                                                                                                                                                                                                         | AB-10105    | Adrian Barton      | 14355.610999999997 | 5             |
|                                                                                                                                                                                                                                                                                                                                         | SC-20095    | Sanjit Chand       | 14142.333999999999 | 6             |
|                                                                                                                                                                                                                                                                                                                                         | KL-16645    | Ken Lonsdale       | 14071.917          | 7             |
|                                                                                                                                                                                                                                                                                                                                         | HL-15040    | Hunter Lopez       | 12873.297999999999 | 8             |
|                                                                                                                                                                                                                                                                                                                                         | SE-20110    | Sanjit Engle       | 12209.438000000002 | 9             |
|                                                                                                                                                                                                                                                                                                                                         | CC-12370    | Christopher Conant | 12129.072          | 10            |
|                                                                                                                                                                                                                                                                                                                                         | TS-21370    | Todd Sumrall       | 11885.871          | 11            |
|                                                                                                                                                                                                                                                                                                                                         | GT-14710    | Greg Tran          | 11820.119999999997 | 12            |
|                                                                                                                                                                                                                                                                                                                                         | BM-11140    | Becky Martin       | 11609.900000000001 | 13            |
|                                                                                                                                                                                                                                                                                                                                         | SV-20365    | Seth Vernon        | 11420.645999999997 | 14            |
|                                                                                                                                                                                                                                                                                                                                         | CJ-12010    | Caroline Jumper    | 11079.742          | 15            |
|                                                                                                                                                                                                                                                                                                                                         | CL-12565    | Clay Ludtke        | 10880.545999999998 | 16            |
|                                                                                                                                                                                                                                                                                                                                         | ME-17320    | Maria Etezadi      | 10635.917999999998 | 17            |
|                                                                                                                                                                                                                                                                                                                                         | KF-16285    | Karen Ferguson     | 10604.266          | 18            |
|                                                                                                                                                                                                                                                                                                                                         | BS-11365    | Bill Shonely       | 10501.652999999998 | 19            |
|                                                                                                                                                                                                                                                                                                                                         | EH-13765    | Edward Hooks       | 10005.980000000001 | 20            |
|                                                                                                                                                                                                                                                                                                                                         | JL-15835    | John Lee           | 9403.003           | 21            |
|                                                                                                                                                                                                                                                                                                                                         | GT-14635    | Grant Thornton     | 9351.211999999998  | 22            |
|                                                                                                                                                                                                                                                                                                                                         | HW-14935    | Helen Wasserman    | 9300.253999999997  | 23            |
|                                                                                                                                                                                                                                                                                                                                         | TB-21400    | Tom Boeckenhauer   | 9012.741999999998  | 24            |
|                                                                                                                                                                                                                                                                                                                                         | PF-19120    | Peter Fuller       | 8980.496           | 25            |

## Insight

This provides a complete customer performance report.



**Step 7: Solve business queries (top customers, low customers, single-order customers, above-average sales).**

## TOP CUSTOMERS

```
SELECT
    `Customer ID`,
    SUM(Sales) AS Total_Sales
FROM orders
GROUP BY `Customer ID`
ORDER BY Total_Sales DESC
LIMIT 10;
```

## Output

| Result Grid     Filter Rows: <input type="text"/> |             |                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|
|                                                                                                                                                                                                                         | Customer ID | Total_Sales        |
| ▶                                                                                                                                                                                                                       | SM-20320    | 25043.05           |
|                                                                                                                                                                                                                         | TC-20980    | 19017.847999999998 |
|                                                                                                                                                                                                                         | RB-19360    | 15117.339          |
|                                                                                                                                                                                                                         | TA-21385    | 14595.62           |
|                                                                                                                                                                                                                         | AB-10105    | 14355.610999999997 |
|                                                                                                                                                                                                                         | SC-20095    | 14142.333999999999 |
|                                                                                                                                                                                                                         | KL-16645    | 14071.917          |
|                                                                                                                                                                                                                         | HL-15040    | 12873.297999999999 |
|                                                                                                                                                                                                                         | SE-20110    | 12209.438000000002 |
|                                                                                                                                                                                                                         | CC-12370    | 12129.072          |

Insight:

These customers generate the highest revenue.

## LOW CUSTOMERS

```
SELECT
    `Customer ID`,
    SUM(Sales) AS Total_Sales
FROM orders
GROUP BY `Customer ID`
ORDER BY Total_Sales ASC
LIMIT 5;
```

### Output

| Result Grid    Filter Rows: <input type="text"/> |             |             |
|-----------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|
|                                                                                                                                   | Customer ID | Total_Sales |
| ▶                                                                                                                                 | TS-21085    | 4.833       |
|                                                                                                                                   | LD-16855    | 5.304       |
|                                                                                                                                   | MG-18205    | 12.32       |
|                                                                                                                                   | CJ-11875    | 16.52       |
|                                                                                                                                   | RS-19870    | 22.328      |

### Insight:

These customers may require retention strategies

### Single – order Customers

```
SELECT
    `Customer ID`,
    COUNT(`Order ID`) AS Order_Count
FROM orders
GROUP BY `Customer ID`
HAVING COUNT(`Order ID`) = 1;
```

### Output

| Result Grid |             |             | Filter Rows: |
|-------------|-------------|-------------|--------------|
|             | Customer ID | Order_Count |              |
| ▶           | JR-15700    | 1           |              |
|             | MG-18205    | 1           |              |
|             | AO-10810    | 1           |              |
|             | LD-16855    | 1           |              |
|             | CJ-11875    | 1           |              |
|             | RE-19405    | 1           |              |

Insight:

These customers may require retention strategies

## Above – Average Customers

```

WITH customer_sales AS
(
    SELECT
        `Customer ID`,
        SUM(Sales) AS Total_Sales
    FROM orders
    GROUP BY `Customer ID`
)
SELECT *
FROM customer_sales
WHERE Total_Sales >
(
    SELECT AVG(Total_Sales)
    FROM customer_sales
);

```

## Output

| Result Grid |             |                    | Filter Rows: | Export: | Wrap |
|-------------|-------------|--------------------|--------------|---------|------|
|             | Customer ID | Total_Sales        |              |         |      |
| ▶           | BH-11710    | 6255.351000000001  |              |         |      |
|             | IM-15070    | 4930.473999999999  |              |         |      |
|             | PK-19075    | 8158.654           |              |         |      |
|             | TB-21520    | 4730.628           |              |         |      |
|             | MA-17560    | 4280.620999999999  |              |         |      |
|             | SN-20710    | 3127.9592000000007 |              |         |      |
|             | LC-16930    | 4481.162           |              |         |      |
|             | RA-19885    | 3832.3139999999994 |              |         |      |
|             | ES-14080    | 4648.516           |              |         |      |
|             | KM-16720    | 4909.472000000001  |              |         |      |
|             | KD-16270    | 8282.358           |              |         |      |
|             | JE-15745    | 6400.454999999999  |              |         |      |
|             | SC-20770    | 4492.656999999999  |              |         |      |
|             | CS-12400    | 3651.8585000000007 |              |         |      |
|             | GM-14455    | 3078.6240000000003 |              |         |      |
|             | RB-19465    | 4556.566000000002  |              |         |      |
|             | GZ-14470    | 4355.15            |              |         |      |
|             | JM-15250    | 3145.3740000000003 |              |         |      |

Insight:

These are the highest-value customers compared to the average customer.

## Highest Order value per customers

```
SELECT
    `Customer ID`,
    MAX(Sales) AS Highest_Order_Value
FROM orders
GROUP BY `Customer ID`;
```

## Output

| Result Grid     Filter Rows: <input type="text"/> |             |                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------|
|                                                                                                                                                                                                                     | Customer ID | Highest_Order_Value |
| ▶                                                                                                                                                                                                                   | CG-12520    | 731.94              |
|                                                                                                                                                                                                                     | DV-13045    | 721.875             |
|                                                                                                                                                                                                                     | SO-20335    | 957.5775            |
|                                                                                                                                                                                                                     | BH-11710    | 1706.184            |
|                                                                                                                                                                                                                     | AA-10480    | 479.97              |
|                                                                                                                                                                                                                     | IM-15070    | 1819.86             |
|                                                                                                                                                                                                                     | HP-14815    | 1242.9              |
|                                                                                                                                                                                                                     | PK-19075    | 4305.552            |
|                                                                                                                                                                                                                     | AG-10270    | 1336.44             |
|                                                                                                                                                                                                                     | ZD-21925    | 823.96              |
|                                                                                                                                                                                                                     | KB-16585    | 1013.832            |
|                                                                                                                                                                                                                     | SF-20065    | 254.9               |
|                                                                                                                                                                                                                     | EB-13870    | 1044.63             |
|                                                                                                                                                                                                                     | EH-13945    | 100.704             |
|                                                                                                                                                                                                                     | TB-21520    | 3083.43             |
|                                                                                                                                                                                                                     | MA-17560    | 799.984             |
|                                                                                                                                                                                                                     | GH-14485    | 1097.544            |
|                                                                                                                                                                                                                     | SN-20710    | 532.3992            |
|                                                                                                                                                                                                                     | LC-16930    | 2396.4              |
|                                                                                                                                                                                                                     | RA-19885    | 1215.92             |
|                                                                                                                                                                                                                     | ES-14080    | 1319.8              |
|                                                                                                                                                                                                                     | ON-18715    | 630.024             |
|                                                                                                                                                                                                                     | PO-18865    | 604.752             |
|                                                                                                                                                                                                                     | LH-16900    | 638.82              |

## Insight

It identifies the largest purchase made by every customer and helps understand customer spending behavior.

